RAM and Storage Theory

1.Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3	DDR4	DDR5
Data Transfer Speed	800–2133 MT/s	1600–3200 MT/s	4800–6400+ MT/s
Voltage	1.5V	1.2V	1.1V
Performance	Lower	Better than DDR3	Faster than DDR4
Power Consumption	Higher	Lower than DDR3	Lower than DDR4
Typical Use Cases	Older desktops and laptops	Modern desktops and laptops	New gaming PCs and workstations
Pin Count (Desktop RAM)	240 pins	288 pins	288 pins
Memory Architecture	Single 64-bit channel	Single 64-bit channel	Two independent 32-bit subchannels
Compatibility	Supports DDR3 motherboards	Supports DDR4 motherboards	Supports DDR5 motherboards

**2.Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps. **

Step 1: Locate the RAM Compartment

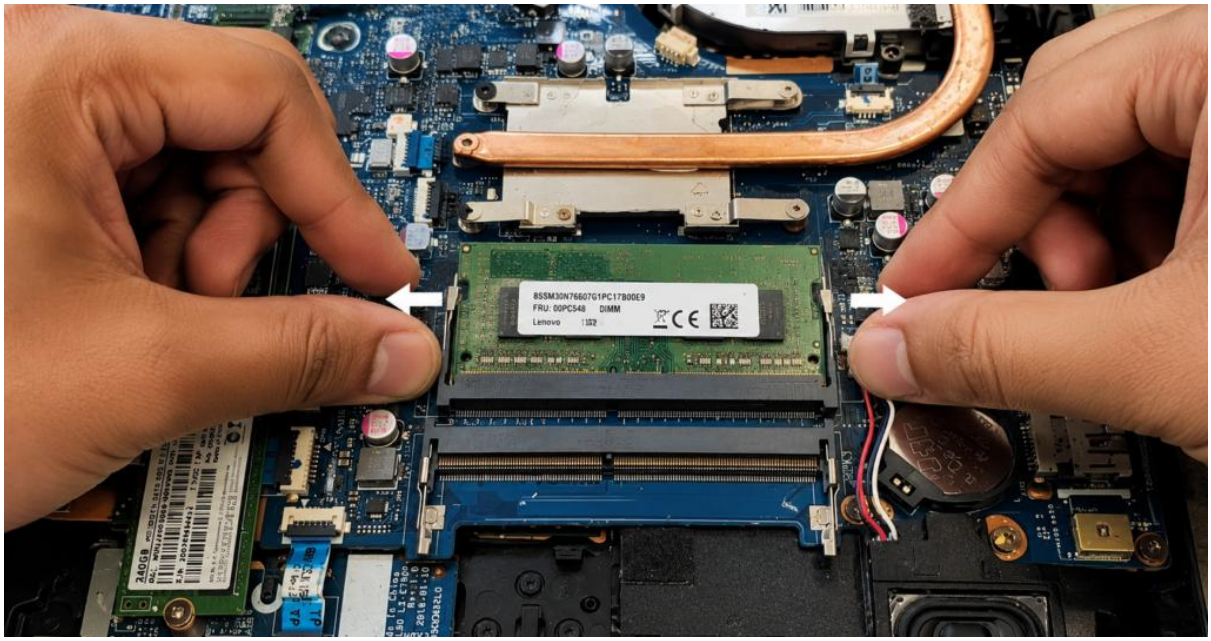
Most laptops have a small panel on the underside, usually held by a few screws, that opens to reveal the RAM and storage slots. On many models, you'll see two types of slots next to each other — one for storage SSD and one or two for RAM, often color-coded green or blue.



Step 2: Remove the Existing RAM (If Upgrading)

If there's already a RAM stick installed and you're replacing or adding to it, look for the small retention clips on either side of the slot. Gently push both clips outward, away from the stick, at the same

time. The RAM will pop up at an angle on its own — simply slide it out from there.

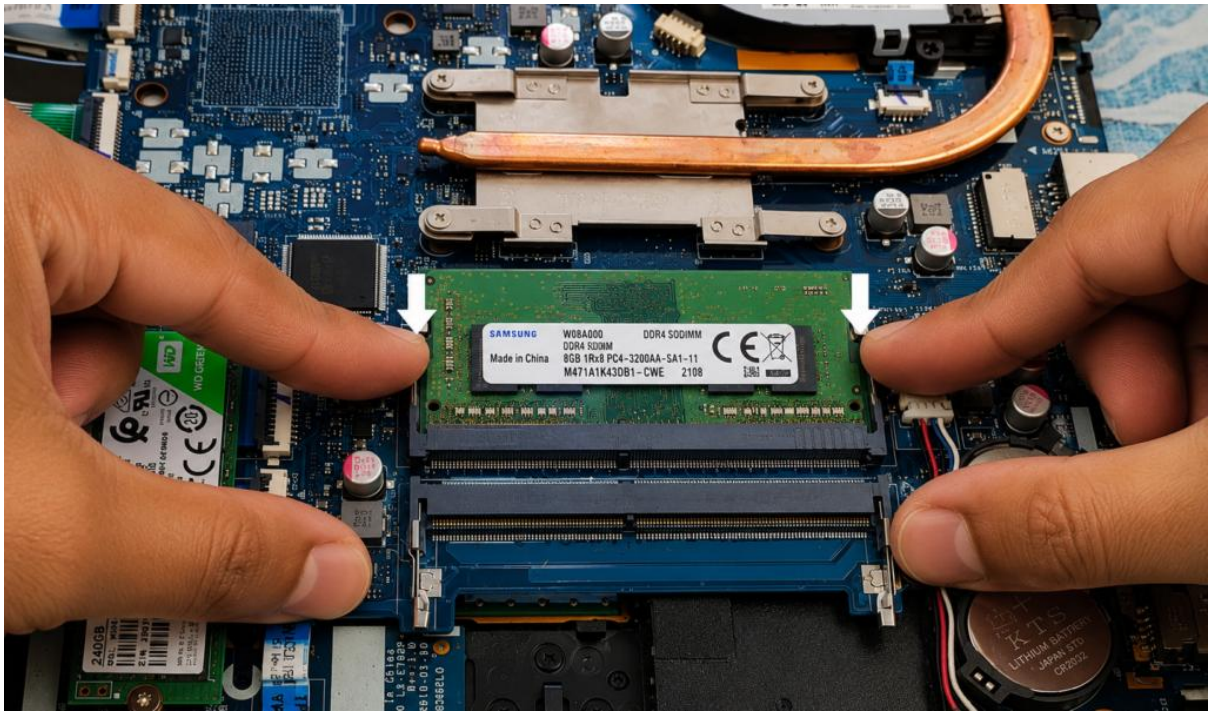


Step 3: Align the New RAM Stick

Every RAM stick has a small notch along its gold connector edge that only lines up one way with the slot — this prevents it from being inserted backward. Hold the stick at roughly a 30-degree angle and line up the notch with the matching gap in the slot.

Step 4: Insert and Press Down

Push the stick firmly into the slot at that angle until it’s fully seated, then press it straight down toward the motherboard. You’ll feel and hear both side clips click into place automatically — this is how you know it’s correctly installed. If it doesn’t click, don’t force it further; remove it, recheck the alignment, and try again.



Step 5: Close Up and Verify

Replace the compartment panel and screws, reinsert the battery if you removed it, and power on your laptop. To confirm the new RAM is recognized, go to Settings > System > About on Windows and check the installed RAM amount matches what you expect.

3.List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Feature	HDD	SSD (SATA)	NVMe SSD
Technology	Uses rotating magnetic disks	Uses flash memory	Uses flash memory with NVMe protocol
Speed	Slow	Faster than HDD	Faster than SATA SSD

Cost	**Cheapest**	**More expensive than HDD**	**Generally more expensive than HDD**
Moving Parts	**Has moving parts**	**No moving parts**	**No moving parts**
Durability	**More sensitive to physical shocks**	**More durable**	**More durable**
Best Use Case	**Storing large files, photos and videos**	**Running Windows and daily applications**	**Gaming, video editing and heavy applications**
Example App / Use	**Google Photos – storing backups and photos**	**Microsoft Word – opening and saving documents**	**Video editing software – loading and editing large video files**

**4.Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model. **

I sincerely apologize for not performing this practical. I did not have permission to open my personal HP laptop. Opening it may damage its internal components or affect its warranty. Therefore, I avoided

removing the storage drive to keep my device safe. Sorry for the inconvenience, sir.